



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI RANGANATHAR & CO

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

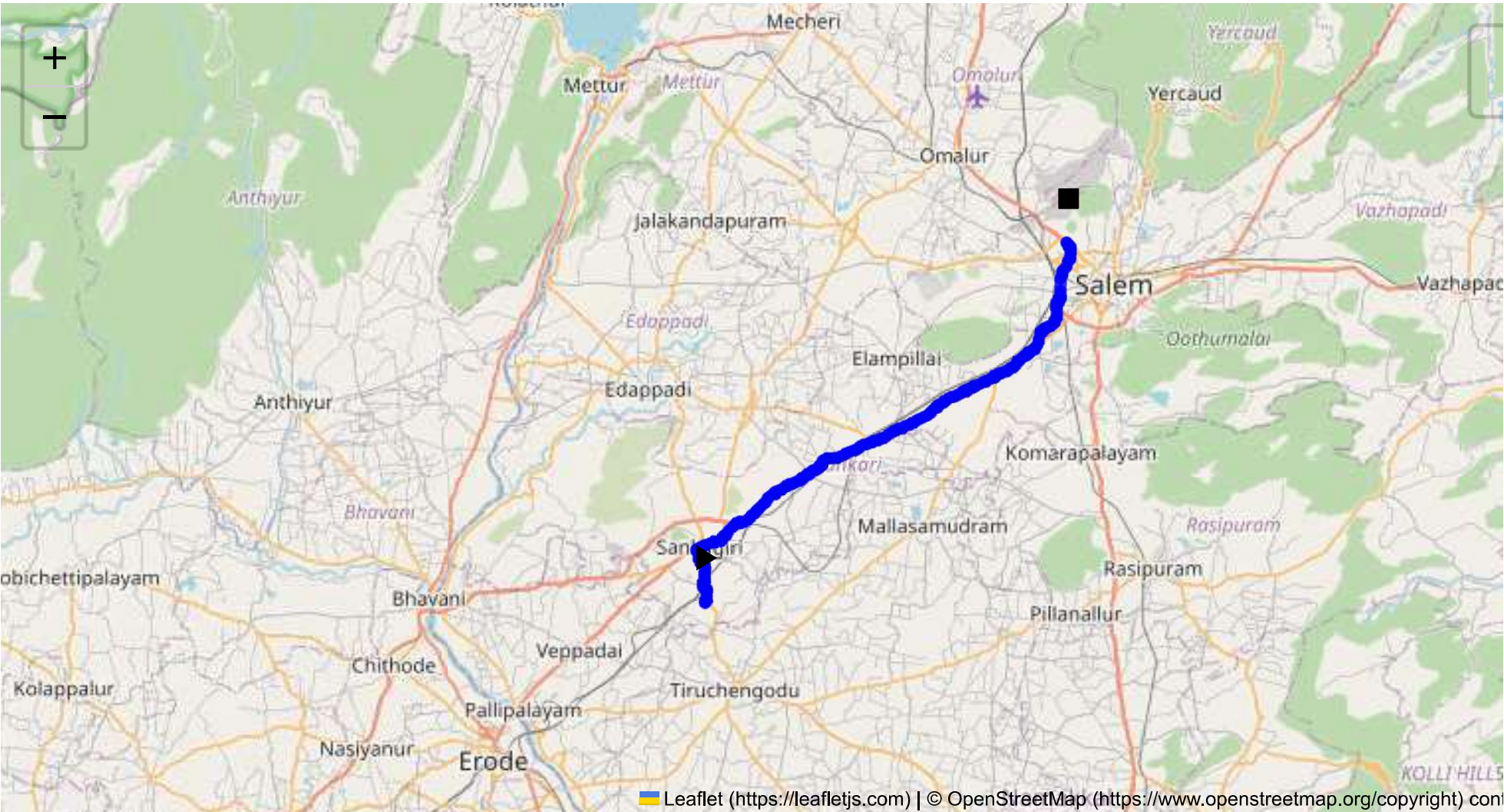
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 43.97 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.682857, 78.12455)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Salem spans approximately 44 kilometers, predominantly following NH 44, one of Tamil Nadu's major highways. This route is straightforward and involves a southeast trajectory passing through both rural and semi-urban areas within Tamil Nadu.

Typical Weather Conditions

Tamil Nadu experiences a tropical climate, with the route affected by both the northeast and southwest monsoons. The region can expect heavy rainfall from October to December due to the northeast monsoon. These conditions can lead to waterlogging, reduced visibility, and slippery road surfaces. During peak summer months, high temperatures might impact vehicle performance, especially concerning engine overheating in heavy vehicles.

Traffic Patterns

The highway experiences significant traffic, especially during morning (8-10 AM) and evening (5-7 PM) peak hours, corresponding with local commuter patterns. Congestion is common around Narasothipatti, approaching Salem, due to urban traffic influx. Expect delays during local festivals and market days.

Road Quality and Infrastructure

National Highway 44 is generally well-maintained, though local stretches may have potholes or uneven surfaces due to heavy rains. Occasional construction or maintenance work could disrupt traffic flow, warranting caution. Adequate signage and lighting are available, yet vigilance is required, particularly during nighttime and rainy conditions.

Alternative Routes for Emergencies

In an emergency, a feasible alternative is to use the state highways connecting through Omalur, though this may add distance and time. It is advisable to have GPS navigation to guide through lesser-known roads or potential detours.

Local Regulations Affecting Hazardous Material Transport

For transport of hazardous materials, permits are mandatory, with restrictions on times of travel in urban localities typically between 6 AM and 11 PM. Compliance with local transportation regulations and safety standards is critical.

Historical Incidents

While NH 44 is considered safe, historical data reveal occasional accidents involving heavy vehicles, primarily due to overspeeding or poor visibility during rains. Fortunately, incidents involving hazardous materials have been minimal. Preventive training and adherence to speed limits are recommended.

Environmental Considerations and Sensitive Areas

The area contains agricultural lands and small water bodies, which should be traversed carefully to prevent environmental contamination. Ensure spill prevention measures are in place for hazardous materials.

Communication Coverage

Mobile network coverage is generally robust along the highway but can be patchy in rural segments. It is advised to check signal availability with multiple service providers if relying on mobile communication.

Emergency Response Times

Emergency services along this route can vary; urban regions around Salem offer quicker response times, approximately 15-20 minutes. Rural stretches may experience delays up to 30-40 minutes. Familiarity with local emergency contacts and ensuring that vehicle telematics systems are active can expedite assistance.

Overall Risk Assessment Summary

The Sangagiri to Salem route via NH 44 is moderately safe for heavy vehicles carrying hazardous materials given adherence to regulations. The primary risks include weather-related hazards, traffic congestion, and communication blackouts in isolated rural stretches. Comprehensive route planning, continuous monitoring of weather conditions, and adherence to safety protocols significantly mitigate these risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45007, 77.87201	15 KM/Hr
5	Turn	High	11.47403, 77.86841	15 KM/Hr
6	Turn	Medium	11.47410, 77.86845	30 KM/Hr
7	Turn	Medium	11.48366, 77.88781	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.61736, 78.10588	30 KM/Hr
11	Turn	Medium	11.61753, 78.10608	30 KM/Hr
12	Turn	Medium	11.64007, 78.11794	30 KM/Hr
13	Turn	Medium	11.64020, 78.11786	30 KM/Hr
14	Turn	Medium	11.64497, 78.12028	30 KM/Hr
15	Turn	Medium	11.64503, 78.12034	30 KM/Hr
16	Turn	High	11.67074, 78.12668	15 KM/Hr
17	Turn	High	11.67081, 78.12661	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
18	Blind Spot	Blind Spot	11.68318, 78.12489	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
23	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
24	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
25	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
27	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
28	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
29	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
26	college	Sona College	11.6768, 78.1257052	30 km/h	Medium

Route Photos of Risky Spots



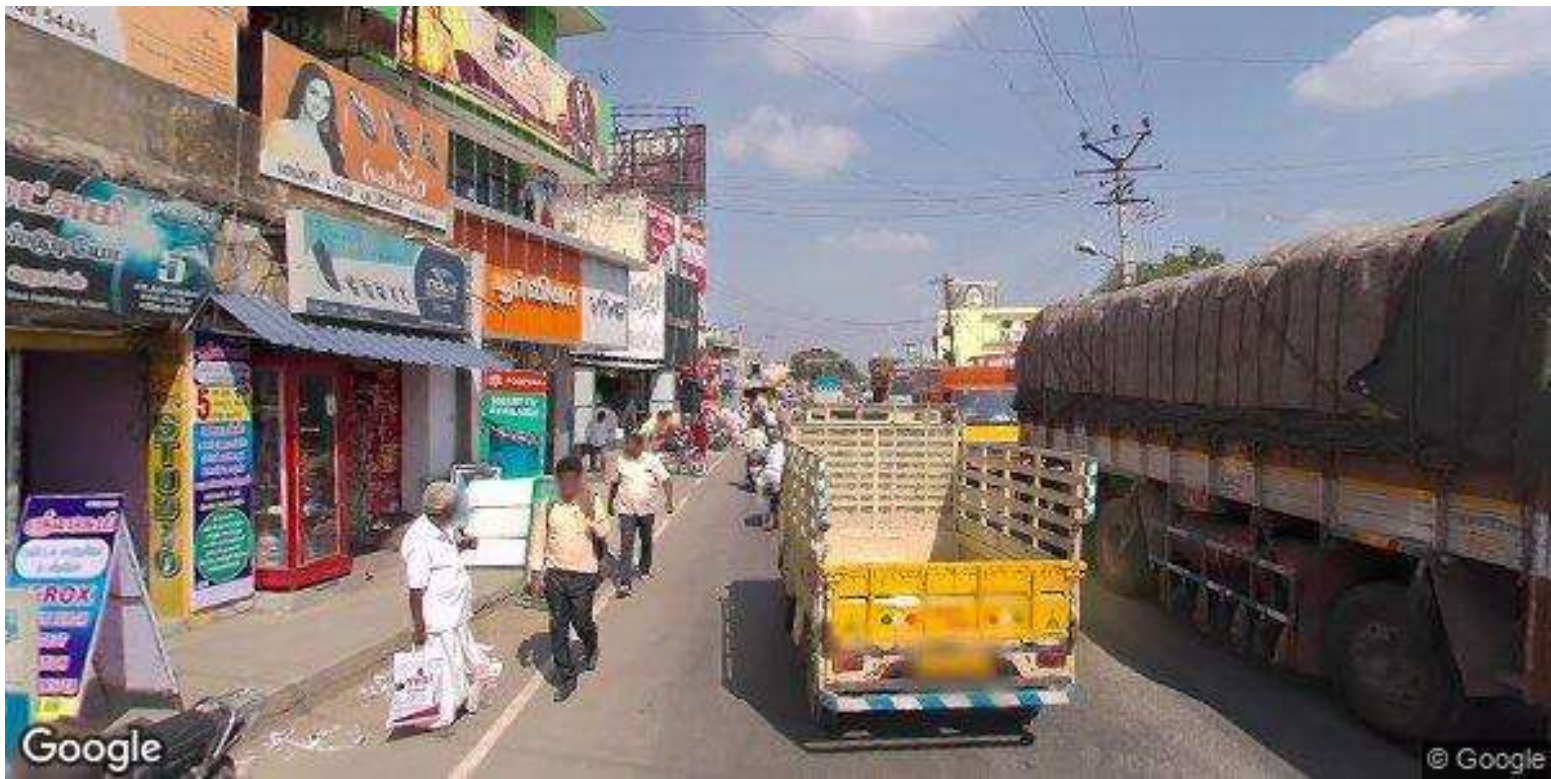
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45007, 77.87201



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47410, 77.86845



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48366, 77.88781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61753, 78.10608



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64007, 78.11794



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64020, 78.11786



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64497, 78.12028



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64503, 78.12034



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67074, 78.12668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67081, 78.12661



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.68318, 78.12489
